

ION

Intelligent Operating Network

AI-augmented Security Operations portal — one platform across the SOC stack, built for air-gapped environments.

v0.38.0

Capability & Architecture Overview

SOC teams drown in tools and alerts



Alert overload

Analysts triage thousands of low-context alerts daily; real threats hide in the noise.



Tool sprawl

Detection, intel, forensics, identity and case data live in separate consoles that don't talk.



Slow, inconsistent triage

Verdict quality swings with analyst experience and fatigue; MTTR suffers.



Air-gapped reality

High-assurance estates can't reach cloud AI or live feeds — most modern tooling assumes they can.

ION unifies the SOC stack behind one portal — and puts a local AI analyst inside the perimeter.

One portal for the whole SOC workflow

ION (Intelligent Operating Network) is a server-rendered Security Operations portal that ingests alerts, manages cases, drives investigations and response, and embeds a local LLM analyst — all without leaving an air-gapped network.



One platform Alerts → cases → forensics → reporting



Air-gapped first No live external feeds; runs fully offline



AI analyst (Bob) Local Ollama LLM, RAG-grounded



No SPA FastAPI + Jinja2, server-rendered HTML

11+

integrations across the SOC stack

80+

service modules, 60+ API routers

6-layer

RAG prompt stack behind every AI verdict

0 / 0 / 0 / 0

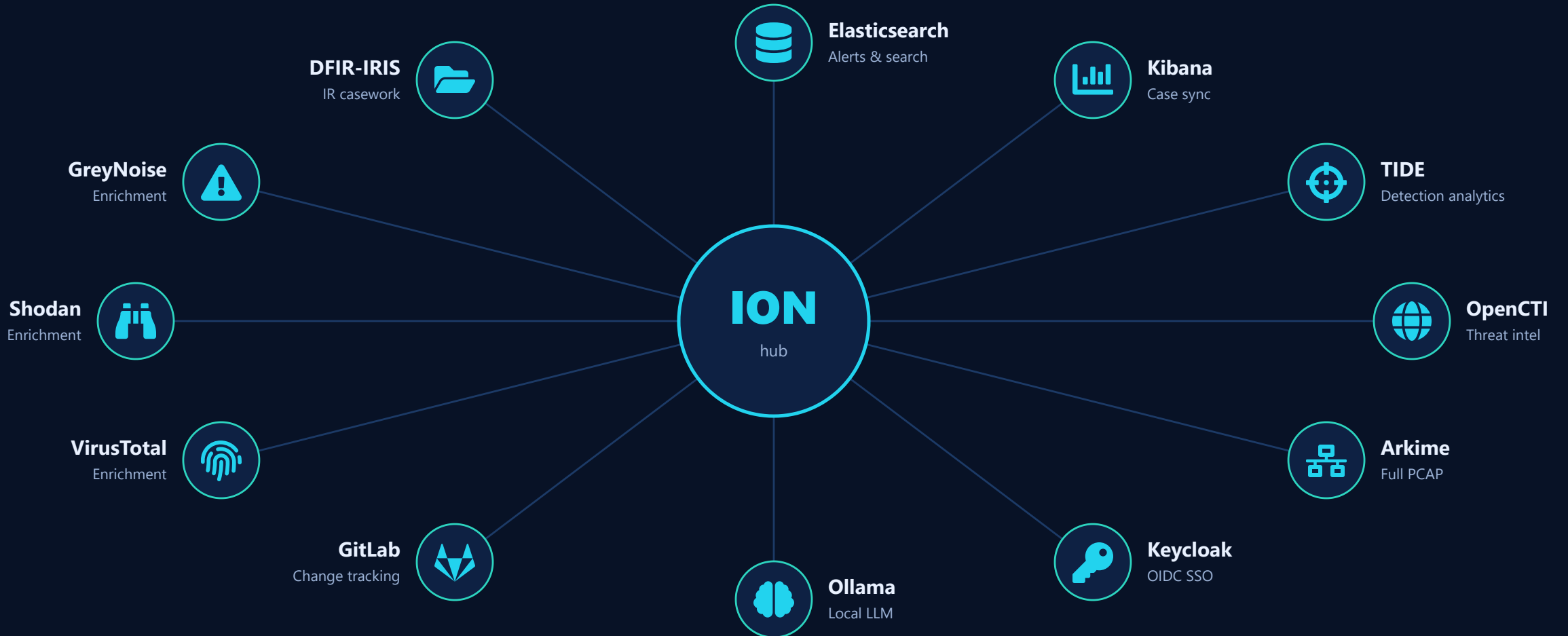
critical / high / med / low findings (v0.38.0)

The alert lifecycle, end to end

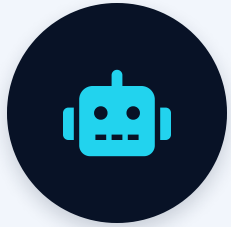


Every closure feeds the next triage — and analyst-agreed cases become AI few-shot exemplars.

11+ integrations, one operating picture



Meet Bob — a SOC analyst that never sleeps



Bob

Local LLM service user



Auto-investigates alerts

Pulls a verdict + reasoning for queued alerts via a per-rule prompt, KB grounding and similar past cases.



Suggests triage

Recommends FP/TP from historical closure patterns before an analyst even opens the alert.



Writes briefings

Daily threat & ops briefings, shift handovers, and case summaries — drafted, analyst-approved.



Stays inside the wire

Runs on a local Ollama model; no alert content ever leaves the air-gapped network.

Outcomes for the SOC



Faster MTTR

Auto-investigation and one-pane workflow cut the time from alert to verdict.



Consistent verdicts

Every alert gets the same RAG-grounded analysis, regardless of who's on shift.



Works air-gapped

Full capability — AI included — with zero external dependencies.



Learns from analysts

Agreed closures become exemplars; quality compounds over time.



Secure by design

20-principle posture; tamper-evident ledgers; 0 open findings.



No tool-hopping

Detection, intel, forensics and identity unified behind one portal.

01

Under the Hood

Architecture, integrations and the AI retrieval engine.

System architecture



Browser Tailwind UI · server-rendered HTML · vanilla JS (no SPA)



ION Application

FastAPI · Jinja2 · SQLAlchemy — 4 uvicorn workers, 1 background leader

RBAC + OIDC

60+ API routers

Bob RAG engine

Advisory-lock loops



PostgreSQL 16

primary store + pgvector (HNSW)



Detection & Intel

Elasticsearch · Kibana · TIDE · OpenCTI



Forensics & Identity

Arkime · Keycloak · GitLab · DFIR-IRIS



Local AI

Ollama (nomic-embed-text + chat)

Find and triage threats



Elasticsearch

alerts · search

- Poll security alerts by index pattern
- ECS-normalised triage state
- Discover / hunting queries



Kibana

bi-directional cases

- Two-way case status & assignee sync
- Space-aware detection scoping
- Comments mirrored both ways



TIDE

detection analytics

- Rule analytics & quality scores
- MITRE ATT&CK coverage & gaps
- DuckDB-backed, API-keyed

Elasticsearch feeds alerts in; Kibana keeps cases in lock-step; TIDE measures detection coverage.

Add context to every indicator



OpenCTI

threat intelligence

- Actors, campaigns, IOCs, reports
- GraphQL API client
- TTP & report drill-down



Enrichment providers

VT · Shodan · GreyNoise

- Reputation & exposure lookups
- AbuseIPDB confidence scoring
- Per-observable orchestration



Observables & timelines

IOC lifecycle

- TLP/PAP-marked indicators
- Cross-source entity timelines
- Attack-story correlation

Air-gapped note: enrichment providers are integrated where reachable; ION never assumes a live internet path.

Investigate deeper, sign in once



Arkime

full packet capture

- PCAP retrieval by community ID
- Keycloak client_credentials auth
- Traffic analytics dashboards



Keycloak

identity & SSO

- OIDC single sign-on (RS256)
- User federation & auto-provision
- Brokers Arkime OAuth2



GitLab · DFIR-IRIS

change & IR casework

- GitLab change tracking
- DFIR-IRIS IR case handoff
- Forensic chain-of-custody

Bob's 6-layer RAG prompt stack

1

Base security persona ION's analyst system prompt

static

2

Per-rule investigation guide AlertPromptTemplate — 5-tier matcher, MITRE-scoped checklist

authored

3

KB RAG context Top-3 Knowledge-Base articles by cosine ≥ 0.65

retrieved

4

Gold exemplars Past analyst-agreed closed cases (few-shot)

learned

5

Elastic Agent Skills Optional published skill bundles

optional

6

Output contract CaseClosureReason JSON — never budget-dropped

pinned

Token-budget aware

The prompt is capped at ~3,800 tokens for the 8K-context local model. RAG layers are admitted by priority and dropped lowest-first when space runs out — but the output contract is always kept, so Bob never loses its JSON envelope.

Grounded, not hallucinated

Every verdict is anchored to ION's own KB, the matching rule guide, and real prior cases — so the reasoning cites the estate's knowledge, not the model's guesswork.

Vector retrieval, measured



pgvector + HNSW

Case & KB vectors indexed for fast cosine similarity in Postgres.



nomic-embed-text

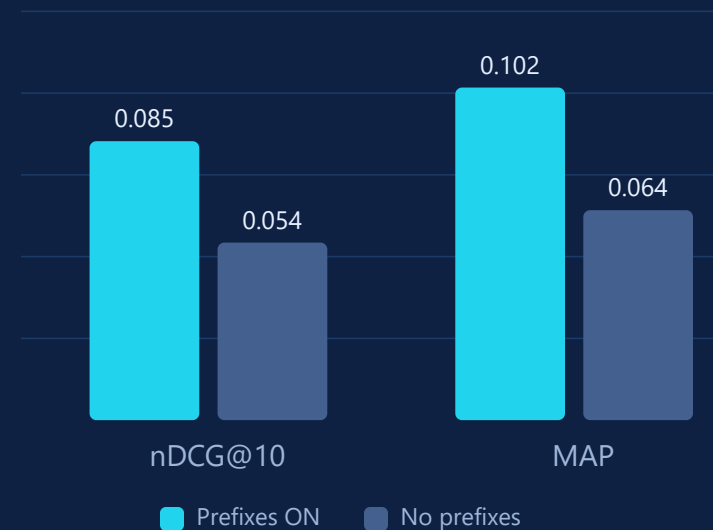
768-dim local embeddings via Ollama — fully air-gapped.



Task prefixes (v0.38.0)

search_query: / search_document: — the model's asymmetric retrieval mode.

v0.38.0 prefixes vs none — ION's real KB



+57% nDCG · +58% MAP — measured on 80 MITRE-tagged KB docs, 43 queries.

02

Trust & Operations

Security model, assurance posture, and running unattended.

Role-based access, scoped by design

10 built-in roles

Analyst tier

analyst · senior · principal

Engineering tier

soc · senior · platform engineer

Specialist & mgmt

forensic · lead

Admin

all permissions



Focus Mode

Hold many roles; activate one at a time to shrink the active permission set.



OIDC SSO

Keycloak (RS256) with optional auto-provisioning; session or bearer auth.



resource:action perms

Fine-grained gates — e.g. alert:triage, case:close, forensic:create.



Defence-in-depth

CSP, HSTS, rate limiting, failed-login monitoring, structured audit logs.

Secure by design — and kept that way

19/20

principles Met

0 / 0 / 0 / 0

C / H / M / L findings

sha256

tamper-evident ledger

per-release

security assessment



Ownership checks before mutation

Auth verified inside the service, not at the edge — no TOCTOU gap.



Tamper-evident workbench

Evidence pins chained with sha256; the ledger detects any change.



Advisory-lock leader election

Exactly-one background execution across workers; safe restarts.



Air-gap as a control

No external feeds shrinks the attack surface by construction.



Tested invariants

Known gotchas pinned by regression tests so they can't silently return.



Documented posture

20-principle audit refreshed every release in SECURITY_ASSESSMENT.

Built to run unattended

Background loops

Kibana case sync	60s	Status & assignee, bi-directional
TIDE data sync	5 min	Rule analytics → Postgres snapshots
Analytics engine	15 min	Risk scoring, repeat offenders, noise
Investigation loop	15 min	Bob processes the auto-investigation queue
Case grouper	10 min	Clusters related alerts into cases
Network mapper	30 min	Asset discovery from Elasticsearch

**Single-leader execution**

pg advisory locks ensure exactly one worker runs each loop.

**Circuit breakers**

External calls fast-fail and auto-recover on outage.

**Crash-safe locks**

A dropped connection auto-releases the lock for re-election.

Analytics, reporting & SOC management



Analytics engine

6 automated jobs: risk scoring, repeat offenders, rule-noise, case metrics.



AI briefings & handovers

Daily threat/ops briefings and end-of-shift reports, drafted by Bob.



SOC health scorecard

5-dimension maturity view; analyst MTTR and FP-rate tracking.



Compliance mapping

Map controls across NIST, ISO 27001/27035, PCI-DSS, SOC-CMM and more.



Executive reports

Scheduled PDF/HTML reports for leadership, generated server-side.



Playbook execution

Step-by-step response with approval-gated automated actions.

Scales from a laptop to a full estate



Minimal

Development

- ION (1 worker)
- Elasticsearch
- SQLite or Postgres



Standard

Production

- Postgres 16
- ION (4 workers)
- Kibana · Keycloak
- Ollama (optional)



Full stack

Air-gapped estate

- + TIDE · OpenCTI
- + Arkime · GitLab
- + Nginx (TLS)
- Full ELK cluster

Containerised with Docker Compose · external stacks join over the ION network · the AI profile adds local Ollama.

The stack, at a glance



FastAPI

Async Python API + page layer



Jinja2

Server-rendered HTML — no SPA



PostgreSQL 16

Primary store



pgvector + HNSW

Similarity search



Ollama

Local LLM + embeddings



Tailwind CSS

Design system



Docker Compose

Packaging & deploy



Keycloak / OIDC

Identity (RS256)

The Bob RAG roadmap



SHIPPED

Prompt-assembly hardening Token-budget guard; output contract protected (v0.35)



SHIPPED

RAG layers default-on Case similarity, KB grounding, exemplars on by default (v0.36–37)



SHIPPED

Asymmetric task prefixes nomic search_query/document — +57% nDCG (v0.38)



NEXT

Playbook & TI-report RAG New embedding surfaces; KB chunking



PLANNED

Memory rework Confidence-sorted recall; analyst-disagreement surfacing

ION

The whole SOC, one intelligent portal.

- Unifies 11+ tools — detection, intel, forensics, identity — behind one server-rendered portal
- A local, RAG-grounded AI analyst that runs entirely inside the air gap
- Secure by design: 19/20 principles met, tamper-evident ledgers, zero open findings